QUIETWATER / PROJECT DOSSIER

Quietwater

Comprehensive Project Report

The gentlest response that works

Technical, product, design, economic, and operational account

Prepared 17 October 2026 from the local Quietwater working tree

CURRENT LOCAL WORKING TREE | 17 OCTOBER 2026

Simulation and decision-support documentation; not an operational authorization.

1. Scope and evidence standard

This report describes the current local project, the problem it addresses, the decisions it makes, the interface a harbor watch officer sees, its simulator, its cost and staffing estimates, its tests, and the work required before an operational deployment. The local working tree contains changes that may not yet be on GitHub; this report therefore describes the inspected files, not an asserted release. The source of truth for current behavior is the executable code and configuration. The README, demo script, backend notes, pipeline audit, and older four-page project handoff were used as context, but their older promises are identified where the implementation differs.

"Working" in this report means the local software runs on simulated contacts. It does not mean a harbor, Coast Guard unit, or research lab has approved or deployed it. Detection, radio and acoustic contact, hold requests, dispatch, and the external notice outbox are mocked. The decision engine, simulator, live updates, handoff workflow, audit receipts, configuration changes, and replay are software features that exist in the project.

2. Executive account

Quietwater is a decision-support tool for a harbor watch officer facing an unidentified autonomous underwater vehicle (AUV/UUV) in a working harbor. It sits between a sonar contact being seen and a person deciding what to do. Its design question is not merely "Is this dangerous?" but "What is the smallest response that handles the evidence and the consequence?" This matters because a port has expensive, safety-critical operations: ships, fuel transfers, submerged cables, divers, traffic lanes, and research activity. Treating every sonar return as an emergency creates needless interruptions and costly shutdowns; ignoring every return misses genuine hazards.

The system classifies a contact, checks identity and schedule, attempts an appropriate contact path, assesses behavior and proximity, and chooses a rung on a six-step ladder: R0 Watch, R1 Look again, R2 Ask, R3 Slow, R4 Hold, R5 Hand off. R0-R3 can be automated under the default rules. R4 and R5 always require a human workflow. The machine drafts an external notice but never transmits it automatically. A second look is capped at two. A hard trigger, such as an improvised-device signature or a contact closing on an asset, goes directly to urgent handoff. Each decision records a plain-language reason, a checklist of what has been ruled out, and why a neighboring rung was not selected.

The interface is deliberately operator-centered: a sweeping harbor radar, Alert Budget, prominent handoff sheet, decision receipts, a live Twist Console, and a collapsible Watch efficiency panel. The same engine also has a recovery mode for the quiet-day problem of finding a lost research vehicle. Configuration profiles illustrate how a local registry, contact channels, notice recipient, language, response windows, and approver labels can change by port without rewriting the ladder. These are demonstration profiles, not verified operational plans.

In a fresh headless run of the checked-out code, the scripted 200-contact flood produced 200 seen, 198 handled quietly, 2 that ever needed a person, and 0 cloud AI calls. It completed 240 simulated seconds in 4.42 seconds of wall time on this machine, with an engine-reported mean of 0.09 ms per decision. Those are reproducible demonstration observations, not field performance or a service-level guarantee. The cost model estimated 396 minutes of manual triage capacity freed under its configured two-minute-per-quiet-contact assumption; that is an opportunity value, not payroll saved.

3. Origin, audience, and boundaries

Quietwater was built for the InnovateLA at USC Dual-Use Hackathon, Marine track with NULL Systems. The

challenge asks what belongs between "seen" and "handled" when an unauthorized autonomous vehicle appears in a working harbor. Its themes are a gentle response, cost and scale, a useful everyday second customer, an adaptable response to live twists, and a reliable five-minute demonstration. Physical interception, jamming, disabling, and capture are explicitly outside the product boundary.

The primary user is a harbor watch officer, not the vehicle operator. The primary institutional setting is a port or waterside security team. The second customer story is a research lab or survey organization looking for its own missing underwater vehicle on an ordinary day. In both modes, the watch officer needs context quickly: what the contact may be, what evidence supports or weakens that reading, what is nearby, which step is recommended, what happens if approved, and whether the machine has already done anything.

The central product promise is proportionality. Cheap automated review happens before scarce human attention. A contact far from anything consequential gets watched or examined again. A contact near working assets may justify a hail or a traffic advisory. Stopping work and sending a notice are separated and require a person. A new port should change a reviewed local profile and thresholds rather than rewrite the engine. None of those choices turns Quietwater into a surveillance or enforcement authority; local governance remains necessary.

4. End-to-end system flow

The data path is: scripted or seeded simulated sonar contact -> pluggable classifier -> derived facts -> deterministic rules engine -> narrow advisor for ambiguous cases -> decision receipt and, if needed, a handoff item -> live user interface. An operator action returns through the API to the handoff state machine and audit log. Configuration from YAML informs the classifier mapping, thresholds, geography, consequences, ladder text, response windows, jurisdictions, optimization parameters, and simulator behavior. The same Session class runs a live scenario and a replay preview, which makes before/after rule comparisons meaningful.

The classifier interface accepts an observation and returns a label, confidence, score map, and source. The current MockClassifier uses scripted labels where provided and otherwise derives a deterministic noisy label from scenario truth and the number of second looks. That truth is internal to the simulator, not evidence the live decision engine would have. The adapter is clear, but a pretrained sonar model has not been integrated. Fine labels include research AUV, commercial ROV, military UUV, improvised device, biologic, debris, and unknown; the coarse scan label compresses these into marine life, debris, known vehicle, or unknown for display and routing.

The facts stage adds registry status, scheduled time, authorized zone, class/size/speed/signature consistency, nearest active asset, distance-based consequence, projected closest approach, trend, approach direction, evasion after a hail, nearby traffic, and whether a compatible communication route exists. The final Decision record contains the selected rung or rungs, urgency, response window, one-line reason, path through the tree, ruled-out checks, neighboring-rung explanations, contact facts, recommended action text, and any advisor result. The hub writes receipts and audit entries and pushes snapshots over Server-Sent Events (SSE) to the React UI.

5. Response ladder: exactly what each rung means

- R0 Watch. Log a confident non-vehicle or an identity-consistent, scheduled contact. A known contact in a critical zone may get an informational notice while remaining at Watch. No person is interrupted by default.
- R1 Look again. Schedule another pass from the existing sensor. The default interval is 15 simulated seconds, with at most two second looks. If the contact remains unidentified, shift up one rung; the loop cannot run forever.

- R2 Ask. Request an ID using a compatible acoustic link to the submerged vehicle or radio contact with its support vessel/operator. If no compatible vehicle link exists, the recommendation seeks an operator through port traffic instead of falsely recording a failed vehicle hail. The actual transmissions are mocked.
- R3 Slow. Advise nearby traffic, using named nearby vessels when known. The default security wording says to slow to 5 knots. The advisory is represented as a decision/action in the demo; no actual VHF broadcast occurs.
- R4 Hold. Recommend pausing one operation, or present an area option when the circumstances justify a broader pause. A person chooses and places the hold request. The terminal response is mocked and has success/failure states.
- R5 Hand off. Build a full package and draft a notice to the local recipient. A person opens, reviews, optionally edits, and explicitly confirms the send. The outbox is simulated; it does not contact an agency.

Rungs can be combined. An unknown high-consequence contact and a military-grade contact normally produce R4 plus R5. The highest rung determines whether the decision needs a person. The configured human_required_from defaults to R4 and can be moved lower for stricter human review, but schema validation does not allow it above R4. This invariant protects the human gate even when a twist changes other rules.

6. The implemented decision tree

The current engine uses S0-S5, not the N1-N7 tree described in an earlier planning handoff. S0 first asks whether the contact is confidently fish or debris. The non-vehicle confidence threshold is 0.75 by default; a confident result goes to R0. Lower-confidence cases continue.

S1 checks identity. A vehicle must be registered, on schedule, and in its authorized zone to go directly to S3. Registry entries specify an ID, operator, vehicle type, size, sonar signature, authorized zones, and time windows. Schedule checks include overnight windows. If any element is missing, S2 considers contacting it.

S2 handles the hail. A successful reply returns to S3. A missing, intermittent, mismatched, or pending reply proceeds to behavior analysis, but the pending reply gets time before non-urgent escalation above R2. A contact marked hail_route: unavailable never gets a fabricated hail timestamp; its checklist says that no compatible link exists and that a reply was not expected. The supported scripted contact routes are acoustic, surface, and unavailable.

S3 checks consistency against registration or class data. Size beyond the configured tolerance, unexpected sonar signature, or speed beyond the class margin can make the evidence inconsistent. A consistent contact earns R0, with an informational notice if it is inside a critical zone. Inconsistent evidence proceeds to behavior/consequence or remains in the unknown branch.

S4 tests hard triggers. An improvised-device classification over its threshold, explicit hostile flag, or a sufficiently fast contact heading toward an active asset produces urgent R5. The approach test uses a 300 m radius, a 30-degree cone toward the asset, and a minimum speed of 0.8 m/s by default. Hard triggers bypass the ordinary waiting logic. Evasion is considered at S5 and also produces urgent R5. Evasion is estimated from turning or a speed jump after a hail, with a scripted evasion flag for scenarios.

S5 combines class evidence with consequence. A confidently classified research or commercial vehicle goes back through consistency; this is not automatic trust if other facts disagree. A confident military classification yields R4 plus R5. An unknown low-consequence contact gets R1 and then a capped escalation. An unknown medium-consequence contact gets R2 if no traffic is nearby, or R3 if traffic needs warning. An unknown high-consequence contact gets R4 plus R5. When a hail is pending, the ordinary branch is capped at R2 until the

ten-second timeout, unless S4 has a hard trigger. An urgent R5 can latch so a later quieter snapshot does not immediately erase the urgent handoff.

The code records why the chosen rung is proportionate. "Confidence alone never escalates" means a confidently detected but unidentified contact far from assets does not automatically summon an officer. The relationship between identity and consequence does the work: uncertainty becomes more important when an asset, operation, diver area, or plausible approach path raises the stakes.

7. Consequence, motion, and identity evidence

The default consequence bands are configurable by asset kind. High/medium distances are 200/450 m for fuel docks, 120/350 m for ships, 60/200 m for cables, and 150/400 m for dive areas. The generic fallback is 120/350 m. Only active assets are considered. Geometries include rectangles, circles, and lines with width. The software computes the closest distance to each shape, picks the most consequential active asset, and retains the nearest asset for explanation.

Closest point of approach (CPA) is a look-ahead, not a trained forecast. At the current speed and heading, the engine projects a straight-line path for 120 seconds in five-second increments and checks proximity along it. It may raise the consequence level, capped at medium under the default configuration. This allows a vehicle still distant from a cable or berth to be treated more carefully if it is heading toward it. The trend stage reads recent trail history over a 30-second window and labels closing, leaving, loitering, steady, or new. A contact can change consequence as it moves, and the engine re-evaluates it on subsequent ticks.

Identity uses the active jurisdiction profile's registry. Local profiles can replace the base entries. A registry match alone is insufficient: schedule and authorized-zone conditions must also be met. The communication state is separate from identity. AIS can help with a surface vessel's identity, but it is not a radio conversation with a submerged vehicle. This distinction is represented in the demo contact route and in the wording of the recommended Ask action.

The limitations matter. CPA assumes a straight line even when a simulated vehicle later follows waypoints. Evasion is a heuristic and can misread a sharp turn around land. A time or zone check can be wrong if local data is stale. The current program has local example registries, not a live allied/external registry integration. It has no physical sensor confidence calibration against real harbor data.

8. The narrow advisor and the cost of AI

Quietwater's rules engine is the decision source of truth. An advisor layer only sees ambiguous contacts: unknowns, certain low-confidence vehicles, or anomalies such as an implausibly large "fish." It is excluded from hard-trigger outcomes and urgent cases. The advisor can request a second look or raise the rung by at most the configured max_raise (one by default); it cannot lower a rung. Advice and clamp results appear in receipts.

The default advisor is deterministic and offline. Its score is a weighted sum of compounding signals: low confidence, silence or mismatch after a hail, closing on an asset in the projected path, loitering, implausible size/speed, class inconsistency, and medium consequence. The default raise threshold is 0.6; a lower threshold may request another look. Mere low confidence has a limited weight and does not by itself dictate a costly handoff. The advisor is invoked only when the ambiguity gate admits a contact, which is the system's main compute-cost control.

An optional LLM adapter exists behind configuration and is off by default. It asks for strict structured output, validates its fields, caps its effect, uses a timeout, and caches responses by contact snapshot and configuration hash.

If the model fails, the rule-based advisor answers; if both fail, the system raises one rung rather than quietly accepting uncertainty. The optional transport uses a configured environment variable for its API key and does not place a key in the repository. Current backend notes say an earlier live LLM trial slowed ticks and raised more contacts than the offline advisor, changing the demo story. It is not appropriate to present that optional model as the normal path or as a measured saving. The measured flood used zero cloud AI calls.

9. Simulator: what is generated and what is measured

The simulator advances contacts across a two-dimensional harbor drawing. It models positions, heading, speed, size, sonar signature, motion style, hail response and delay, transponder/ID, hostility and evasion flags, waypoints, events, spawning and departure, and trails. Motion styles include straight movement, drift with a configured current, wandering, loitering, and waypoint-following. It checks water boundaries and piers. The classifier uses a scripted confidence track or deterministic seeded noise. A second look can raise classification confidence or reveal a scripted change. The sonar feed is simulated; no real sonar or image recognition is involved.

The project includes a Morning watch story, an all-archetypes scenario, a 200-contact flood, a lost-vehicle recovery scenario, a contact-paths scenario that contrasts surface contact with no compatible link, and seeded random security and recovery scenarios. Archetypes include bottle, kelp, fish, known research sub, survey and inspection ROVs, unknown distant and near assets, stray research vehicle, evasive contact, improvised signature, equipment, and military UUV. Scripted events can change a contact's answer or identity as the scenario progresses.

Random generation deliberately varies two separate aspects: a watch-sector radius from 350 to 1,300 m in 50 m steps, and a weighted traffic level. Clear water contains 0-2 contacts, quiet 3-8, working harbor 9-30, busy 31-80, and peak surge 81-200. The configured weights are 2, 4, 5, 3, and 1, respectively; they are demo probabilities rather than measured traffic frequencies. A seed determines the same contact set, clock, and geometry for replay. It randomizes identity variants, location inside reachable water, speed, heading, size, confidence trajectory, hail response, event timing, and some motion. Placement regions are chosen to make low-, medium-, and high-consequence contacts likely under the default rules. If the randomly chosen count is very small, it cannot contain every tier. Changes to asset positions or thresholds may also change those tiers even when the seed is unchanged.

The watch area shown in the model is water inside the chosen radius and the demonstration map, not the entire area of a real port. The software estimates its square kilometers by sampling 50 m grid cells that lie in water and in range. A larger port would need multiple sectors or different geometry. A seed makes a replay comparable before and after a rule twist because random generation reads scenario configuration and the seed, not the live decision thresholds. Live ticks and headless preview use somewhat different time steps, so they are comparable decision replays rather than pixel-identical animations.

The default scripted flood is intentionally a stress demonstration: many contacts are ordinary marine life or debris, with a small number of contacts that deserve a human. It tests whether the attention queue grows far more slowly than the sensor feed. The summary counts contacts that *ever* needed a human separately from those still pending at a given moment. Those measures should not be confused.

10. Handoff workflow and authority boundary

An R4/R5 decision creates a handoff item rather than silently executing the recommended action. The item's state machine includes pending, expired, watching, hold requesting, hold active, hold failed, notice review, notice sent, dismissed, resolved, lifted, and stood down. The backend enumerates legal operator moves and legal system moves;

invalid transitions return an error instead of mutating state. A later severity increase can re-open attention even after a contact was dismissed or watched. An overdue item pages a backup officer on screen only; it does not send an external alert.

For R4, the screen presents "this operation only" and "this entire area." The operation choice pauses only the named activity at the focus asset. The area choice covers every active operation inside the proposed radius. The software recommends area when two or more operations are in range, the contact is urgent, or a high-consequence contact is closing; otherwise it recommends one operation. This is advice, not autonomous authority. The operator explicitly chooses the scope. The mock terminal can confirm a hold after about a second or fail it. An active hold has a timer; when time is up, it remains active until a person lifts it or extends it. The system does not quietly resume operations.

For R5, the draft includes recipient, sender, subject, contact description, estimated position, nearby asset, reason, ruled-out list, automated steps already taken, and recommendation. The security recipient depends on the active jurisdiction; recovery mode sends to the research lab operations desk. Opening the draft is separate from sending it. The operator may edit or cancel review; send requires an explicit confirmation. The current outbox records a simulated send. A hold request does not send the notice, and sending a notice does not itself place a hold. These are separate approvals with separate receipts.

Every handoff transition adds an audit entry and a receipt with time, actor string, item, prior and new status, and rule version/configuration hash. The profile also names the intended hold/handoff approver and security liaison in recommendation and notice text. There is no authenticated identity or role enforcement yet: the current operator name is supplied by the client. The state machine enforces the human action step, but an operational deployment would require login, role checks, and signed approvals. Local rules must determine who is authorized to order a hold or transmit a notice.

11. The interface and design rationale

The frontend is a single-page React/TypeScript application with plain CSS, Vite, and a custom canvas radar. The screen is built around the watch officer's workflow rather than an engineering console. It uses a full-screen harbor sonar view, pulsing sweep, trails, asset zones, and contacts colored by consequence: gray for lower concern, yellow for medium, red for higher concern. A hover popup gives the label, confidence, distance, bearing, speed, heading, rung, and a one-line reason. A click on a human-required contact opens its handoff; a click on a quiet one opens its receipt. An acted-on red contact becomes muted and brightens again if it worsens.

The Alert Budget sits near the top and answers the anti-flood question immediately: how many contacts were seen, how many were handled quietly, and how many need the officer now. The underlying budget also tracks how many ever needed a person, active contacts, holds, sent notices, dismissed/expired/resolved items, per-rung counts, and a weighted attention load. Default weights are R0=0, R1=0, R2=0, R3=1, R4=2, and R5=3, with a nominal capacity of 6. This is a communication model of interruptions, not a certified measure of cognitive workload. The old "bang" comparison is the count of all seen contacts, illustrating the cost of treating every hit as an alarm.

The handoff sheet is the visual centerpiece. It puts the contact and urgency first, then the one-line reason and clear next action. It shows what would happen if a hold is approved, offers the narrow and area scopes, keeps the external notice in a separate review step, and displays a response countdown. The "Details" area provides label and confidence, size, speed and heading, nearby asset, position, ruled-out checklist, and history. "Keep watching" snoozes the item for a configurable two-minute simulated interval; dismiss has a ten-second undo window and does not stop tracking. Urgent new items can open the sheet automatically.

Decision receipts support accountability and explainability. The operator can filter all, automated, or human-required entries, search by contact or reason, and expand "Why not one rung higher or lower?" to see the path, checklist, consequence, and alternative-rung reasoning. Audit export is available in JSON and CSV. A Watching panel collects snoozed contacts. A Dual-use toggle changes security to research-vehicle recovery; the security checks still apply to threats in recovery mode.

The bottom controls provide pause/resume, replay, flood, receipts, twist access, and speed controls. Keyboard shortcuts include Space for pause, R for replay, F for flood, T for Twist Console, and Escape to close a sheet. The scenario generator button produces a seeded new situation. The Watch efficiency panel is collapsible and sits in a corner to keep cost and staffing measures accessible without dominating the radar. Its wording deliberately separates "value freed" from literal money saved.

The visual approach favors immediate recognition: high contrast, large touch targets, short summaries, consistent risk colors, and detail only when expanded. The product is not a command-and-control weapon screen. Its screens stop at human review and simulated communications. There is no neutralize, intercept, or physical-stop control.

12. Live twists and deterministic comparison

The Twist Console exposes editable thresholds, rung mappings, zones, schedules, presets, and YAML. A user can preview a change before applying it: the backend runs the same seeded scenario under the old and proposed rules, then compares contact-by-contact peak outcomes, rung changes, paths, first time a human was requested, and totals for quiet versus human-needed contacts. The UI shows both the changed rule fields and the decision effects. A successful apply persists to config/rules.yaml, hot-reloads, and re-evaluates active contacts. Twist history is saved with before/after information and supports revert.

The standard demo twist is to activate Fuel Dock B and mark its zone critical while Morning watch is running. The new asset changes consequence for a nearby contact; a previous advisory can become Hold plus Hand off. This shows the product's central adaptation argument: the harbor's facts and policy change, while the decision code stays the same. An invalid YAML edit is rejected and the last valid rules remain active. The server polls file modification time during its tick, so changes usually appear quickly, but the mechanism is not a robust production file watcher.

Jurisdiction selection is also configuration-driven. The included examples are Los Angeles, Singapore, and Rotterdam (keyed eu). Profiles set registry source and optionally a profile registry, notice recipient, surface and acoustic contact guidance, port operations/security channels, language, approval-role labels, and response windows. The same ladder can therefore be replayed against different local data. This is a portability demonstration, not proof that one port's radio plan, security duties, or legal approval process automatically applies elsewhere. The map remains the Los Angeles demonstration geometry even when a Singapore or Rotterdam profile is selected. Profile entries and recipients require local signoff. A Port Facility Security Officer is a distinct role under a local security plan; the ladder itself is a product design, not a requirement claimed to come from the ISPS Code.

13. Quiet Tuesday: the second customer

Recovery mode addresses a research organization's missing vehicle. The configured example is USC Wrigley Marine Lab's REMUS-02. A possible match may be identified from transponder, sonar signature, size and class, and confidence. The system uses capped re-scans for uncertain matches, then pings or recommends a recovery handoff when the vehicle is confirmed. It estimates drift heading and speed, projected position over ten minutes, and approximate entry into a critical zone. It can recommend notifying traffic and, where there is an immediate operational

hazard, a hold. The handoff package goes to the lab's operations desk instead of a security recipient.

The second customer matters commercially and ethically. A harbor does not encounter a hostile AUV every day, but labs and survey teams can lose equipment during ordinary work. The same sensor feed and rules machinery can reduce search time and protect traffic while recovering a legitimate vehicle. However, a profile-specific registry currently replaces the base registry; under Singapore or Rotterdam, the configured USC lost vehicle may not be present. That recovery story therefore works as configured in the Los Angeles profile, and each new port needs its own recovery target and registry entries.

14. Optimization model: objectives, inputs, and formulas

The optimization layer is a scenario-based estimate of three scarce things: officer attention, operations paused, and paid cloud computation. It aims to spend the expensive resource last. The rule engine and offline advisor handle ordinary contacts before a person is called; a second sonar look is capped; and the optional network model remains off unless explicitly enabled. The model is visible in the Watch efficiency panel. It is not an independently trained optimizer, a validated staffing forecast, or an accounting system. It reads actual decisions from the simulated session and combines them with configurable assumptions.

Inputs come from the current run and rules.yaml: watch radius, water area, total seen contacts, first time each contact needed a human, peak rung and response window, active assets and their geometry, current number of officers, modeled review minutes for a hold or handoff, target utilization, standard and candidate area-hold radii, safety buffer, hold duration, officer hourly rate, paused-operation hourly rate, and cost per optional LLM call. Current defaults are one officer, a 60-minute watch horizon, 70% target utilization, 4 minutes to review a hold, 3 minutes to review a handoff, 2 minutes of manual review per quiet contact, $60 per officer-hour, $500 per paused-operation-hour, $0.02 per LLM call, a 400 m standard area-hold radius, candidate radii of 150/250/400 m, and a 25 m buffer. These are demo assumptions, not measured harbor prices or labor contracts.

The contact-count relationship is direct: handled quietly = contacts seen - contacts that ever required a human. The panel's "officer time freed" is handled quietly x assumed manual minutes per contact. Its dollar capacity value is freed minutes x assumed hourly officer rate / 60. This value means the on-duty officer can spend that time on other work. It is not a claim that the port pays fewer wages during the shift. In particular, the tool should not be presented as having automatically eliminated a position.

The staffing calculation treats each first human request as a task with an arrival time, response deadline, and modeled service duration. It starts with a workload lower bound: ceil(total review minutes / (watch horizon x target utilization)), never below one. It then tries increasing numbers of operators and schedules tasks by arrival and deadline using the earliest available operator. The first count that finishes every modeled review by its deadline is reported as "officers needed." If none within the searched range works, the panel says review takes too long. This is a simple deterministic planning approximation. It does not model breaks, simultaneous radio duties, individual skill, shift overlap, uncertain review duration, or queuing outside the selected scenario. A response-window failure can produce a recommendation for more than one officer even when total review minutes are small.

The hold-radius calculation begins with all active assets already in a high-consequence band of the contact, plus the focus asset even if it is farther away. It takes the greatest distance among those assets and adds the configured safety buffer. It picks the smallest candidate radius meeting that minimum; if all candidates are too small, it rounds the minimum upward to a 50 m increment. The same hold-options logic then lists operations that fall inside the radius and recommends either one operation or the area according to the evidence. A person still selects and approves the actual scope. The baseline comparison uses the standard 400 m area radius. The model multiplies the number of

operations in the recommended option by the configured hold duration to estimate paused operation-minutes in each case.

Costs are computed as follows. Estimated watch cost = modeled staffing cost + modeled paused-operation cost + optional cloud-AI cost. Staffing cost uses the recommended staff count if a feasible one was found, otherwise the configured current count, multiplied by hourly rate and watch-horizon fraction. Pause cost uses operation-minutes multiplied by the paused-operation hourly rate. Cloud cost is LLM calls x configured per-call price; it is zero by default because the offline advisor is selected. Avoided pause cost is the positive difference between baseline and modeled pause costs. The panel's "automation value freed" adds officer capacity value to avoided pause cost. Its manual-review benchmark is estimated cost plus that value, and its percentage is value divided by that benchmark. These are comparative demo quantities, not an audited return-on-investment statement.

To compare differently sized sectors, the panel divides both estimated cost and value freed by the sampled water area in square kilometers. This denominator is important: a $20 modeled benefit over a small cove and a $20 benefit over a large watch sector do not represent the same density of support. The watch radius and water area are both shown. The model does not claim that officer effort scales linearly with area; it uses area as a comparison lens while contact mix and response deadlines drive staffing.

Important caveats are built into the formula. The pause saving can legitimately be $0 when the optimized and standard scopes both recommend the same single operation, or when there is no R4 hold. Dollar savings vary with assumed manual review time and prices. The panel's single "recommended hold radius" is the minimum among modeled holds; the full range is also in backend metrics. A hypothetical value of freed attention should not be added to actual cash savings without a real operating study. The model has no validated false-positive or missed-risk cost, and does not optimize every legal or safety constraint a port would use.

15. Measured demonstration runs

These measurements were generated on 17 October 2026 by loading the current local configuration into the Hub, advancing each scenario one simulated second at a time, and reading its budget and optimization metrics. The environment and implementation affect wall time; figures are useful as a reproducible demo check, not a guarantee for live sensors. No optional LLM was enabled.

- Morning watch (300 simulated seconds): 12 contacts seen; 10 handled quietly; 2 ever needed a person; 2 pending at measurement; 0 cloud AI calls. Final active rung counts were 8 R0, 1 R2, 1 R3, and 2 R5. The watch radius was 1,000 m, with 1.91 square kilometers of sampled water. Under the two-minute manual-review assumption, 20 officer minutes and $20 of capacity value were modeled as freed. The reported estimated watch cost was $620. Headless wall time was 0.48 s; mean measured decision time 0.12 ms.
- Flood (240 simulated seconds): 200 contacts seen; 198 handled quietly and 2 ever needed a person; 2 pending; 0 cloud AI calls. Final active rung counts were 159 R0, 39 R2, and 2 R5. At the 1,000 m radius and 1.91 square kilometers of water, the model counted 396 minutes and $396 of officer-capacity value under its manual-review assumption. It reported $620 estimated watch cost and two modeled officers needed to meet the response windows. Headless wall time was 4.42 s; mean measured decision time 0.09 ms.
- Seeded random harbor, seed 42 (120 simulated seconds): 27 contacts seen; 22 handled quietly; 5 ever needed a person; 0 cloud AI calls. The generated radius was 900 m and sampled water area 1.82 square kilometers. The model counted 44 freed minutes and $44 of capacity value, with an estimated watch cost of $2,300 and a five-officer response-window recommendation under its assumptions. Wall time was 0.42 s; mean measured decision time 0.11 ms. This is a useful reminder that different mixes can create sharply different human demand,

even at a smaller radius.

These counts describe simulated traffic and default rules. The 200-contact flood was designed to demonstrate anti-flooding and should not be treated as a measured real-world alert rate. "Need a person" counts any contact that crossed the human threshold during the run; "pending" counts those still waiting when measured. The final rung distribution is only a snapshot of active contacts and does not tell the whole path each contact took. The optimization display is a model layered over these outcomes, not evidence that a real port saved the quoted dollars.

16. Software architecture, storage, and interfaces

The backend uses Python 3.11+, FastAPI, Pydantic, Uvicorn, and ruamel.yaml. The frontend uses React 18, TypeScript, Vite, custom canvas radar rendering, and plain CSS. Tests use pytest and HTTPX; Ruff checks Python code. Make and Docker compose provide startup paths. make dev starts auto-reloading API and Vite development servers; make build creates the frontend production bundle and byte-compiles the backend; make test runs backend tests, lint, and TypeScript checking; make start builds and serves UI plus API together on port 8000; docker compose up builds and runs the packaged version. Development defaults are API port 8000 and UI port 5173. Python 3.11+, Node 20+, and Make are required for local development. No API key is needed for the default offline path.

The hub owns the live Session, handoff items, receipts, audit log, mock outbox, rule store, twist history, and SSE subscribers. The browser fetches initial state/configuration and then receives snapshots via /api/stream. Operator actions go back over ordinary HTTP. The stream sends a hello snapshot on connection, subsequent updates, and keepalive messages. In production the backend can serve the built SPA. YAML is the editable rule source. Some twist history is persisted to a JSON file; most live state remains in memory.

The API includes read endpoints for health, configuration, current state, rules, receipts, handoff states, and audit export; streaming at /api/stream; simulation control and scenario selection; security/recovery mode switch; handoff action; and twist preset, history, preview, apply, and revert endpoints. Development mode also exposes a mock-hold-failure switch to demonstrate a refused hold. Invalid handoff moves are rejected. A receipt carries a rule version and configuration hash so reviewers can associate a decision with the rules that produced it. JSON and CSV export are available, but the in-memory audit buffer is bounded.

The configuration files have distinct jobs: config/rules.yaml holds the validated decision and optimization policy; config/harbor.yaml holds drawing geometry and origin; config/scenarios.yaml holds archetypes, randomization slots, scripted scenarios, and mock-classifier data; config/twist_presets.yaml holds ready-made live changes. The frontend files in frontend/src/components implement the cards and sheets, while frontend/src/radar handles canvas behavior. The backend's engine.py is the decision logic, facts.py derives context, advisor.py handles narrow advice, optimization.py calculates cost and staffing estimates, handoff.py defines legal item transitions and hold options, hub.py connects decisions to UI state, and twists.py manages comparable replays.

17. Validation and reproducibility

The project has unit and integration tests for the engine's branches, second-look cap, human floor, pending-hail cap, urgent latching, recovery behavior, advisor validation and fallback, scenario determinism, random generated tiers, API handoff transitions, export, hot reload, twist comparison, and cost model behavior. A full make test completed in this local environment with 176 passing tests, Ruff clean, and TypeScript type checking clean. A subsequently added targeted test for the new contact-path scenario also passed. make build completed successfully. One deprecation warning came from the test client's AnyIO/Starlette dependency, not a Quietwater assertion failure. The report's own

measurements were obtained from headless runs described above rather than copied from a pitch claim.

The important design invariants are explicitly tested: a contact's confidence alone does not force an alert; an unanswered hail gets a timeout; R4/R5 are never automated; the second-look loop is bounded; a notice is not sent before human review and confirmation; an invalid transition changes nothing; a twist compares the same seed against old and new rules; and a missing compatible communication link does not fabricate a failed hail. The tests demonstrate software behavior under configured examples. They do not validate sonar performance, port radio procedure, legal authority, cybersecurity, human factors, or real economic impact.

For a live demo, the reliable path is: load Morning watch; show quiet contacts and a handoff; inspect one decision receipt; run the flood to show 200 in and two human-needed; apply a Fuel Dock B twist and show a changed decision; then switch to vehicle recovery. The demonstration script includes a backup video recommendation. The report deliberately avoids repeating older static slide numbers as if they applied to every seed or rule configuration.

18. Global transfer: what is portable and what is local

The portable core is the reasoning pattern: classify, verify, ask when possible, consider consequence and motion, then choose the least disruptive step that still addresses the evidence. A port in another country can preserve that ladder while supplying its own registry, authorized windows and zones, communication protocols, notice recipient, language, response windows, and approval duties. The project illustrates this with Los Angeles, Singapore, and Rotterdam configuration profiles and a before/after replay path. The communication distinction is central: a submerged AUV may require a compatible acoustic link; a surface support vessel/operator can be called on an appropriate VHF channel; AIS is an identity source, not a hail channel. No compatible link is recorded as such, not as refusal.

This is a configuration architecture, not a turnkey international deployment. The included port profiles are examples requiring review by local port and security authorities. The harbor geometry, assets, and randomization placement remain the Los Angeles demonstration map. Local registries are sample data. Recipient and role strings do not authenticate a person or connect to a government system. VHF working channels and language may depend on sector and purpose. Before a pilot in any new port, the operator would need a port-approved data profile, mapping, contact protocol, staffing plan, legal approvals, and user testing. The International Ship and Port Facility Security framework gives a broad security context; it does not prescribe Quietwater's six rungs.

19. Responsible use and failure boundaries

The technology could be misused to monitor legitimate vessels, divers, or research teams, or to give an automated label more authority than it deserves. The current safeguards are to focus the demo on unscheduled autonomous contacts, keep receipts and ruled-out checks, require a person above the automated rungs, separate a hold from an external notice, and stop at a handoff rather than offering physical interference. The rule-based path can work without cloud calls. A communication failure is represented as uncertainty rather than guilt, and a compatible link that never existed is not treated as an ignored hail.

These safeguards are software choices, not a complete operational safety case. A real deployment would need an approved mission scope, a retention and access policy, sensor validation, false-alarm and miss-rate studies, human factors testing, a role-authenticated approval chain, failover behavior, secure interfaces, and procedures for correcting registry errors. Even apparently gentle actions such as a traffic advisory or a lane hold can have safety and economic consequences. The operator, port authority, and relevant public agencies would retain responsibility for those decisions.

The current system never physically halts a vehicle. The terminal hold request, radio/acoustic hails, advisories, recovery dispatch, and external notice send are mocks. The "notice sent" state means a simulated outbox record, not actual transmission. The absence of live sensor input is intentional for the hackathon but must be resolved for a pilot. Model confidence values are scripted or synthetic; they are not calibrated probabilities for a real sonar system.

20. Known weak spots and implementation gaps

The most important gap is between the demo and an operational harbor. The local handoff PDF describes an extended N1-N7 proposal with an allied/external registry fallback, parallel sensitive-zone warnings, diver recall, and special approval for broadcasts naming object type or intent. The checked-out engine implements S0-S5 and does not implement all of those extended nodes as written. It has configurable local/profile registries, a zone critical flag, consequence bands for divers, and general traffic advisory behavior, but it does not perform a live allied registry lookup with timeout, run an independent parallel zone-warning service, or execute an automated diver recall. Those proposal elements should not be claimed in a demo of the current code.

Other material limitations, drawn from the current backend notes and code, are:

- No durable operational record. Most receipts, handoff items, outbox state, and audit entries live in memory and are lost on restart. Twist history and YAML persist. The audit export deque is bounded and older entries fall off.
- No user authentication. The client supplies an operator name. The state machine forces an operator action, but it cannot verify the person's role or authority. This is particularly important for the jurisdiction approval fields.
- Simulated links. No real sonar, acoustic modem, VHF radio, AIS exchange, Coast Guard interface, port terminal hold system, or recovery dispatch has been integrated. A profile string describes a route; it does not execute one.
- LLM latency. Optional uncached model calls can block the simulation tick and change the number and timing of alerts. The offline default is the credible demo path.
- Simplified motion and consequence. CPA projects straight ahead; trails and evasion are heuristic. Dynamic traffic, currents, turns, and crowded infrastructure would need stronger models and evaluation.
- Simplified queue model. Staffing uses fixed modeled review durations and deadlines. It cannot replace a workload study or account for concurrent watch duties and local staffing rules.
- Simplified economics. Rates and manual-review baselines are assumptions. "Capacity value" is not realized cash unless operations are changed and measured.
- Global profiles remain illustrative. Sample recipients/channels and registry entries need local confirmation; the map and randomization regions do not change with the selected jurisdiction. A profile-specific registry may hide the default recovery vehicle.
- Hot reload and preview are demo-grade. File polling can miss very fast changes; concurrent editor and UI writes can race; replay may use a different time step from live animation; history storage grows; and the schema does not fully enforce a monotone ladder.
- Streaming and scale need hardening. Each tick sends a full SSE snapshot, potentially tens of kilobytes for a 200-contact scene. Slow clients drop old messages. There is no event-ID resume, backend worker separation, or production observability.

These are not reasons to discard the prototype. They are the boundary between a reliable hackathon demonstration and a fielded decision-support system. The first production investment should target trustworthy inputs, durable records, verified roles, safer integration, and measured outcomes before adding more AI.

21. Prioritized roadmap toward a pilot

First: validate the problem with a port partner. Obtain representative, permissioned sensor/contact records and a local security/recovery plan. Define which contacts are in scope, what counts as a false alarm or missed hazard, and who has authority to approve each rung. Confirm the actual radio/acoustic paths and external notice recipients. Collect baseline officer workload and pause costs rather than relying on demo assumptions.

Second: harden the evidence and audit layer. Integrate a real classifier through the existing interface; calibrate confidence, evaluate drift in murky/noisy conditions, and show known performance by contact class. Add a durable append-only audit store, authenticated roles, signed approval events, and tamper-aware configuration versions. Keep the rule engine replayable against historical contact snapshots.

Third: put a narrow, observable pilot in a harbor. Start in advisory/shadow mode with no real hold or external send integration. Compare recommended rungs with watch-officer judgments. Measure contacts in, quiet outcomes, human interruptions, missed hazards, false escalations, review minutes, time to decision, cloud calls, operator trust, and cost per watch sector. A real pilot should explain why the alert budget and staffing model are--or are not--accurate.

Fourth: integrate actions only with approved controls. If the pilot earns trust, connect approved radio, registry, and terminal systems through operator-initiated, logged interfaces. Keep the notice send a human action. Add rollback, failover, incident review, and periodic port-profile validation. Any optional LLM should remain an audited, bounded, off-tick advisor, not the primary gate.

22. Venture and funding path

The primary prospective U.S. federal funding pathway is a relevant future DHS Science & Technology SBIR topic. Quietwater would pitch a decision layer above detection, with research milestones for trustworthy classification inputs, proportional routing, human approval, explainability, and measured performance under many simultaneous contacts. DHS has previously explored autonomous vessel tracking for ports and waterways, which is evidence of mission interest, not a currently open Quietwater-specific call. An applicable Navy SBIR topic might be an alternative if it asks for harbor decision support rather than physical delay, denial, or interception. A Defense Innovation Unit Commercial Solutions Opening could support a later prototype if an open solicitation matches the product and a government partner sponsors it.

Customer-side funding is different from startup funding. Port authorities and facility operators may seek FEMA Port Security Grant Program awards to buy eligible security capabilities. The FY2026 opportunity listed $95 million in estimated program funding and closed July 24, 2026. That round is closed; a future round, eligibility, matching obligations, and permissible project costs would have to be checked at the time. The venture cannot assume a grant award or claim that a port will buy the tool. The commercially credible path is to pair a documented port need and pilot partner with a matching solicitation, then produce performance evidence that a buyer can defend.

23. How to run, inspect, and reproduce the report

From the project root, install Python 3.11 or newer, Node 20 or newer, and Make. Run make dev for the API at http://127.0.0.1:8000 and Vite UI at http://127.0.0.1:5173. Use make dev UI_PORT=5174 if the UI port is occupied. make build produces the static bundle; make start serves the production UI and API together on port 8000; make test runs backend tests, Ruff, and TypeScript checking; docker compose up is the container path. The default advisor needs no secret or cloud account. The README contains the complete command reference.

To inspect the behavior, start Morning watch and open an ordinary contact's receipt. Advance until a contact earns a handoff, then compare one-operation and area hold scopes; review the draft notice as a separate step. Replay from the same seed. Activate Fuel Dock B in the Twist Console or YAML and preview the resulting contact-by-contact diff. Run Flood and check the observed 200/198/2/0 result under default rules. Switch to recovery mode to see the same decision framework applied to a lost research vehicle. Select Contact paths to inspect the difference between a surface operator route and a vehicle with no compatible hail link.

24. Terms and interpretation

- AUV/UUV: Autonomous or unmanned underwater vehicle; the demo uses both common terms in security and research settings.
- Contact: A tracked sonar return with an ID, current state, trail, classification, and supporting evidence; it is not proof of a hostile vehicle.
- Rung: One level of the response ladder. A decision may contain more than one rung, such as Hold plus Hand off.
- Consequence: Low, medium, or high impact potential based on active nearby assets and straight-line closest approach, not a prediction of intent.
- CPA: Closest point of approach along a projected path over a configured horizon.
- Hail: An attempted communication using a compatible vehicle acoustic link or its surface operator's radio route; actual transmission is mocked.
- Receipt: The logged explanation for a decision or handoff transition, including rule version and why a neighboring rung was not chosen.
- Alert Budget: Counts quiet handling, pending human items, and weighted attention load; it is an anti-flooding design metric.
- Capacity value: Assumed officer minutes freed by automation expressed at an hourly rate; it is not necessarily cash savings.
- Twist: A configuration change replayed against the same scenario to show how outcomes move without a code rewrite.
- Profile: A jurisdiction-specific bundle of local registry and communication/approval text. Example profiles require local validation.

25. Source inventory and external references

The inspected local sources were README.md, DEMO_SCRIPT.md, BACKEND_NOTES.md, AUDIT_REPORT.md, PIPELINE_AUDIT.md, TECH_STACK.md, config/rules.yaml, config/scenarios.yaml, config/harbor.yaml, config/twist_presets.yaml, backend application and test modules, frontend components/styles/types, and the earlier Quietwater_Handoff.pdf dated 17 October 2026. The report resolves discrepancies in favor of current executable code and states uncertainties rather than turning planned work into implemented work. The headless measurements in Section 15 were run from the local working tree for this report.

Relevant official program/context sources: U.S. Small Business Administration SBIR overview (https://www.sbir.gov/about); DHS's earlier autonomous-vessel-tracking outreach (https://www.dhs.gov/sites/default/files/publications/dhs_sandt_insights_outreach_sbir_july6_2021.pdf); Defense Innovation Unit commercial pathway (https://www.diu.mil/work-with-us); FY2026 Port Security Grant Program notice

(https://www.grants.gov/search-results-detail/362824); IMO ISPS overview (https://www.imo.org/en/ourwork/security/pages/solas-xi-2%20isps%20code.aspx) and AIS overview (https://www.imo.org/en/ourwork/safety/pages/ais.aspx); Singapore Maritime and Port Authority channel information (https://www.mpa.gov.sg/port-marine-ops/arrivals-and-departures/vessels-arriving-in-singapore); and Port of Rotterdam VHF guidance (https://www.portofrotterdam.com/en/contact-harbourmaster/vts-services-and-vhf-communication-procedure). These sources frame the examples; none certify this prototype for operational use.

Appendix A. Default decision and timing reference

These values are defaults in the inspected rules.yaml; a Twist Console edit or another profile can replace them.

- Classification confidence: research/commercial/military class branch at 0.70; confident non-vehicle threshold 0.75; improvised-device hard-trigger threshold 0.60.
- General consequence distance: high within 120 m, medium within 350 m. Asset-specific distances: fuel dock 200/450 m; ship 120/350 m; cable 60/200 m; dive area 150/400 m.
- CPA look-ahead: 120 s, checked every 5 s, with raising capped at medium by default.
- Approach hard trigger: within 300 m, heading within 30 degrees toward an asset, speed at least 0.8 m/s.
- Evasion heuristic: over a 20 s window after hail, at least 120 degrees of total turning or speed at least 1.8 times the reference speed.
- Hail timeout: 10 s. Low-consequence second looks: at most two, 15 s apart, then one rung higher.
- Trend: 30 s window; closing/leaving threshold 0.05 m/s; loitering threshold 75 degrees of turning or speed below 0.03 m/s.
- Identity consistency: size tolerance 0.5 m and speed margin 1.2 times the class maximum.
- Nearby traffic search radius: 400 m. Default urgent response window 300 s, hold-plus-handoff window 1,800 s, and ordinary human window 900 s; local profiles override these.
- Attention budget: capacity 6; weights R0/R1/R2=0, R3=1, R4=2, R5=3. Keep-watching recheck after 120 s.
- Hold request: nominal 1,800 s duration, 10 s dismiss undo, 1 s mocked terminal response, standard area radius 400 m, candidate radii 150/250/400 m, buffer 25 m.
- Recovery: configured lost ID USC-REMUS-02, confirmation confidence 0.75, 600 s drift horizon, 3,600 s human window.
- Offline advisor: enabled with rule backend, maximum one-rung raise, raise score 0.6, second-look score 0.35, overall timeout 3 s. The optional LLM backend is not selected by default.

Appendix B. Default outcome-to-rung map

The tree does not simply assign rungs by color. Under default rules, confident fish/debris and identity-consistent contacts use R0. A known contact in a critical zone stays at R0 with an informational notice. A hard threat or evasion uses urgent R5. Military-grade uses R4+R5. Unknown low uses R1 until its capped loop escalates. Unknown medium uses R2 when traffic is absent, R3 when nearby traffic should be warned. Unknown high uses R4+R5. A pending hail generally caps the non-urgent route at R2 while it has time to answer. These mappings live in YAML and can be previewed or changed, subject to the human-approval floor.

In recovery mode the names and goals differ: R0 Log, R1 Re-scan, R2 Ping, R3 Notify traffic, R4 Hold lane, and R5 Recover. A possible lost vehicle starts with capped re-scans; after confirmation, consequence and traffic determine whether notice, hold, and recovery handoff are combined. Security hard triggers retain priority even when the screen is in recovery mode.

Appendix C. API and file reference

For a developer or reviewer tracing a live demonstration, the important API routes are:

- GET /api/health, /api/config, /api/state, /api/rules: service, scenario, and rule visibility.
- GET /api/stream: live SSE snapshots and keepalives.
- POST /api/sim/control: play, pause, restart, or adjust speed; POST /api/sim/scenario: choose a scripted or seeded random scenario; POST /api/mode: security/recovery switch.
- POST /api/handoff/{item_id}: an operator action, including approve, review, edit, confirm send, dismiss, undo, watch, escalate, lift, resolve, or extend; GET /api/handoff/states: legal workflow states and moves.
- GET /api/receipts and GET /api/audit/export?format=json|csv: explanations and audit download.
- GET /api/twist/presets, GET /api/twist/history, POST /api/twist/preview, POST /api/twist/apply, and POST /api/twist/revert/{entry_id}: change and compare policy.
- POST /api/dev/mock-hold: available only when development mode is enabled to simulate a terminal refusing a hold.

The highest-value source files are backend/app/engine.py for S0-S5, facts.py for derived evidence, rules.py for validated configuration and hot reload, advisor.py for bounded ambiguous-case advice, optimization.py for the economic model, handoff.py and hub.py for human workflow and audit, randomizer.py and simulator.py for scenarios, twists.py for comparisons, and main.py for API routes. In the frontend, App.tsx coordinates state, Radar.tsx and radar/ draw the harbor, and the components named HandoffSheet, BudgetCard, OptimizationPanel, ReceiptsSheet, and TwistConsole implement the principal interface. The current README and backend notes should be read alongside the source when preparing an actual pilot proposal.

Appendix D. Product choices and their reasons

Why a ladder instead of a danger score? A numeric score does not tell the officer what may safely happen next. The ladder converts evidence into progressively more consequential actions, with an explicit boundary between automated steps and human authority. It also makes the "gentlest response" testable: the receipt can explain why one rung lower would be insufficient and one rung higher would be excessive.

Why model consequence as well as classification? A contact's label does not determine its effect on a harbor. The same unknown object far from operations is different from an unknown object beside a fuel transfer or diver area. Distance, projected motion, and operations in progress make the decision situational. Classification confidence is treated as evidence, not as a substitute for consequence or intent.

Why a bounded second look? More sensor evidence is cheap compared with stopping work or paging a person, but an unbounded loop would quietly defer a decision forever. Two looks with a time limit make "look again" a real rung rather than a hiding place for uncertainty.

Why put AI behind rules? A network model adds latency, cost, and output uncertainty. The deterministic rules cover

routine cases and hard safety boundaries. Ambiguous contacts can receive extra review, but advice may only raise the tier or ask for more observation. The default offline advisor keeps the demo and ordinary operations independent of a cloud connection.

Why keep hold and notice separate? Pausing a berth and notifying an external agency are different actions with different authority and consequences. One approval should not silently imply the other. The interface and audit state machine make each step visible and reviewable.

Why show a cost model even though it is not validated? The challenge asks for scale, not simply a pretty map. The panel turns officer attention, pause footprint, and cloud calls into explicit assumptions a port can question. It gives a path to validation while showing how automation could protect capacity. Its labels and this report repeatedly distinguish modeled capacity from realized savings.

Why build a recovery mode? The same capability is more valuable if it solves an everyday problem. Finding a lab's missing vehicle uses the same contact evidence and communication routes but changes the goal and recipient. It makes "dual use" operationally concrete without adding a physical intervention capability.